

Required Tools and Accounts

Remote Desktop (Mac users only)

Access to the virtual machine will be through a remote desktop connection, so you will also need a suitable RDP client installed on your computer.

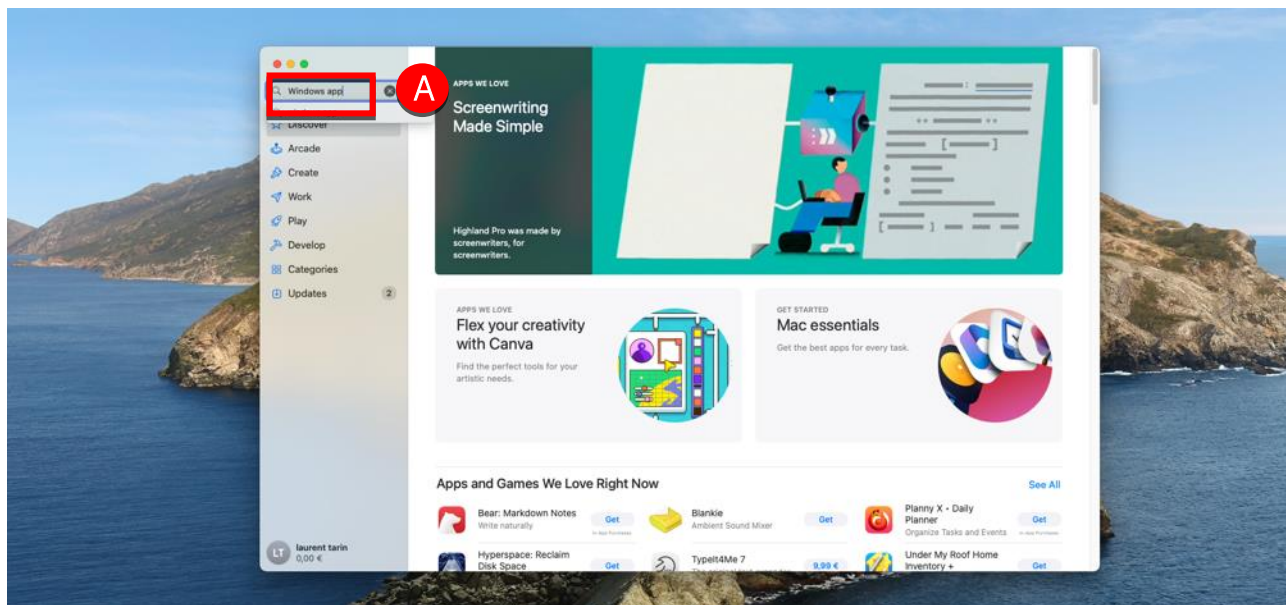
For Windows users, this is already included in the operating system, but for Mac users we recommend you download the Windows App (Microsoft Remote Desktop application) from the AppStore, if you do not already have it installed.

For Mac users, please proceed with the following steps to install Windows App. Windows users can move to the following section.

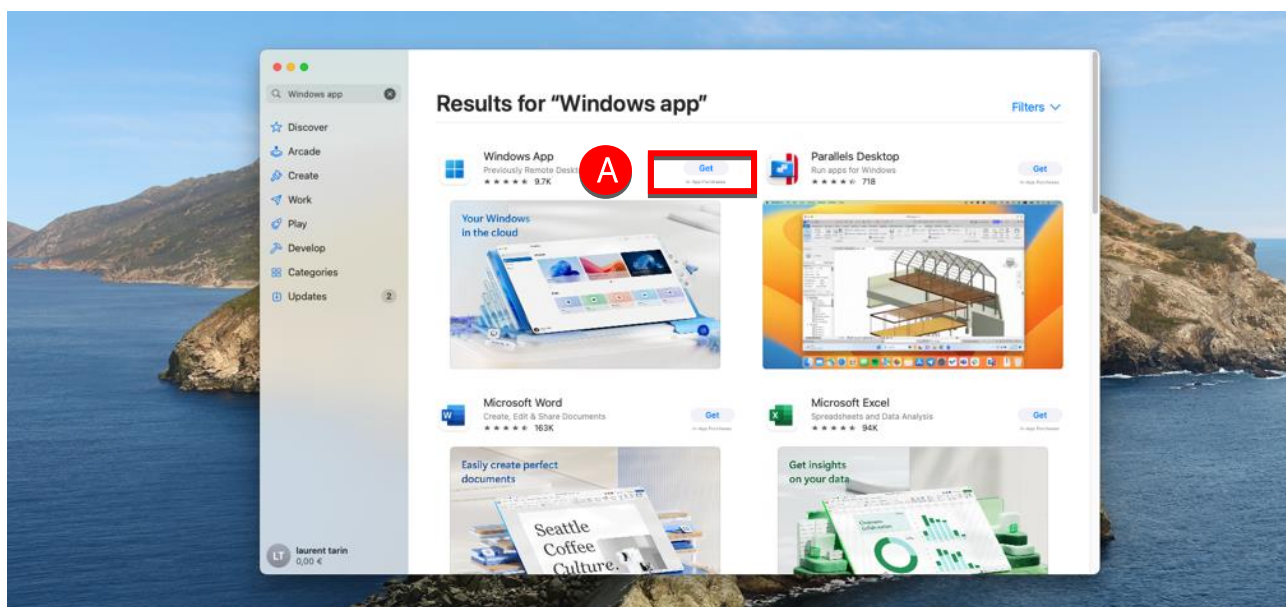
3. Click the **App Store** app icon (A).



4. Type “Windows App” in the search menu (A).



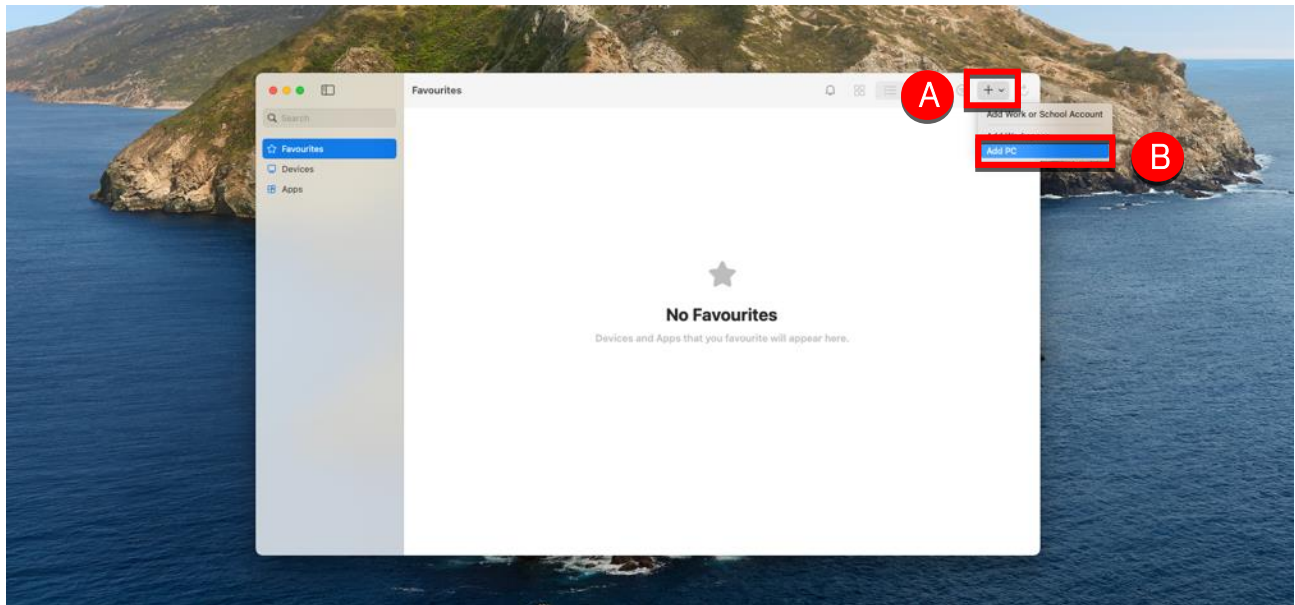
5. Click **Get** next to the Windows App tile (A).



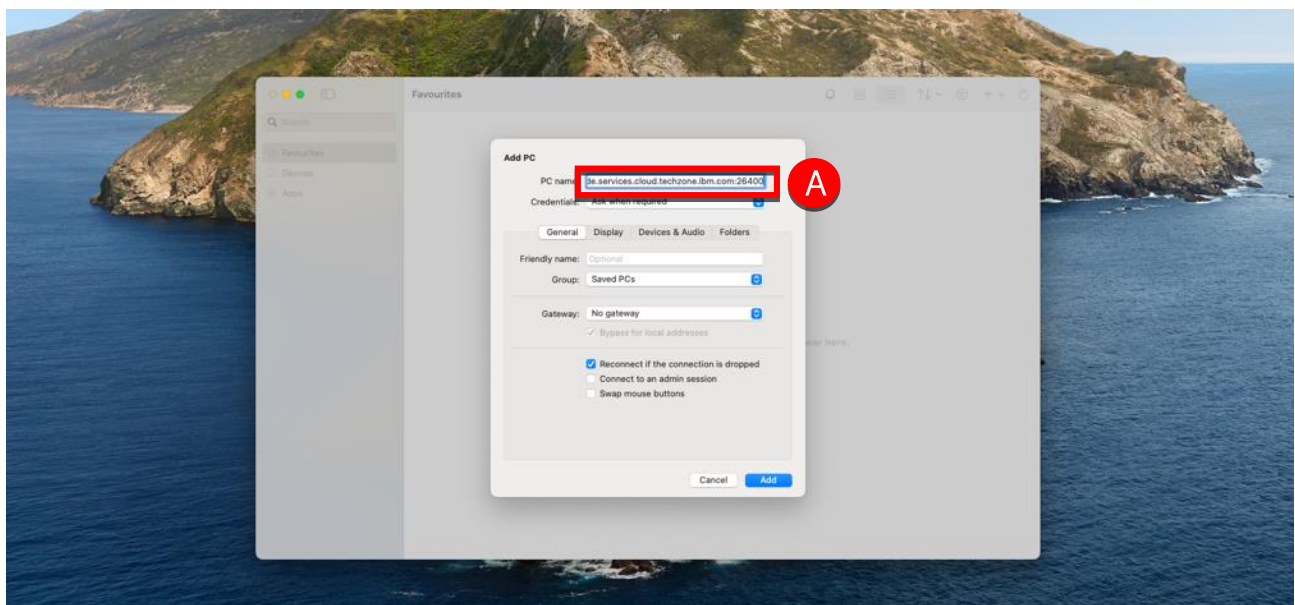
Follow the Windows App installation instructions until the app is installed on your computer.

Windows App to access the VMWare image containing your pre-configured watsonx Orchestrate Developer Edition environment. If you are using Windows, please skip to the next section.

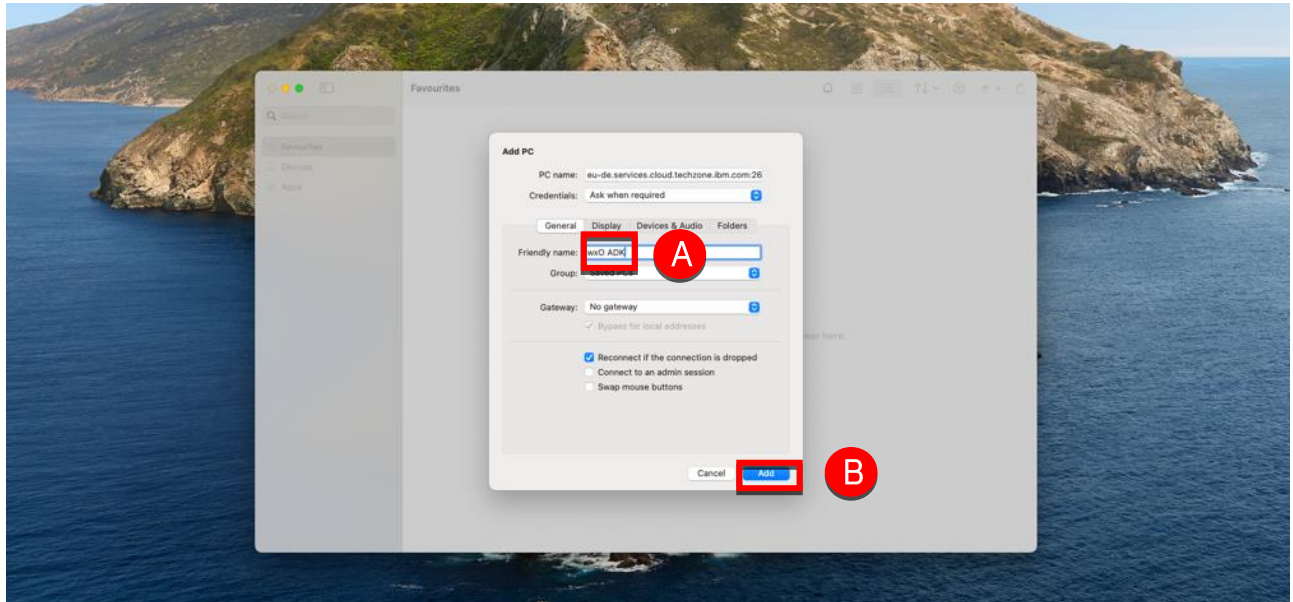
1. Start the Windows App application on your Mac.
2. Click + (A) then select **Add PC** (B).



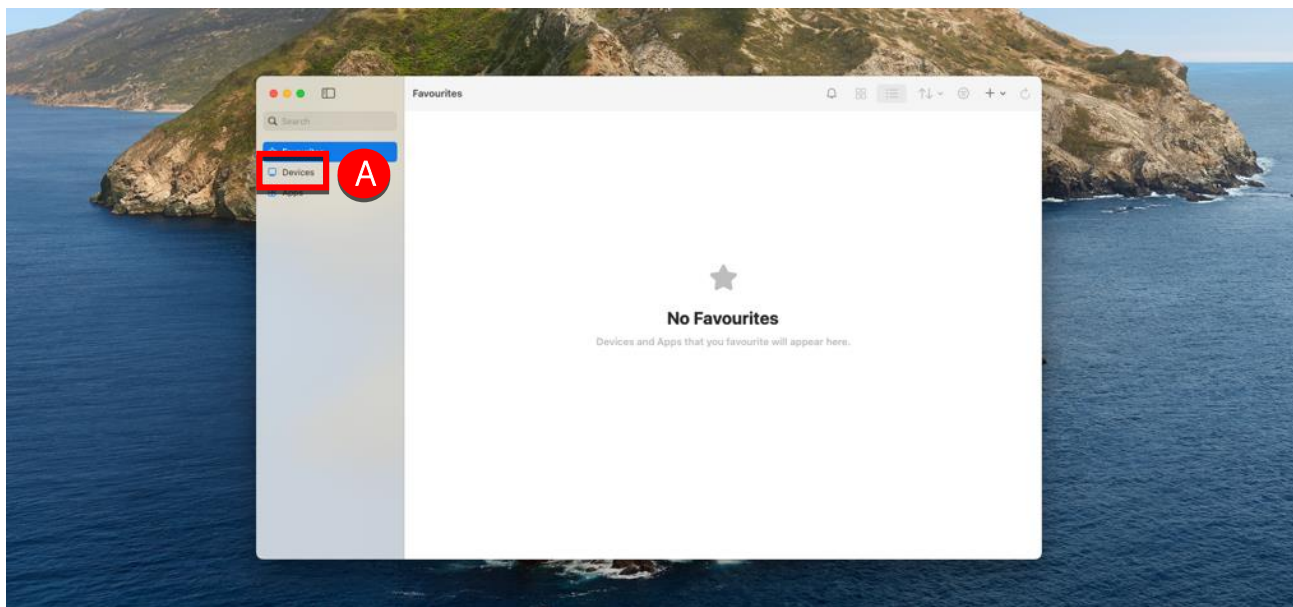
3. Copy your Remote Desktop URL from your lab credentials (item 3) into the **PC name** (A).



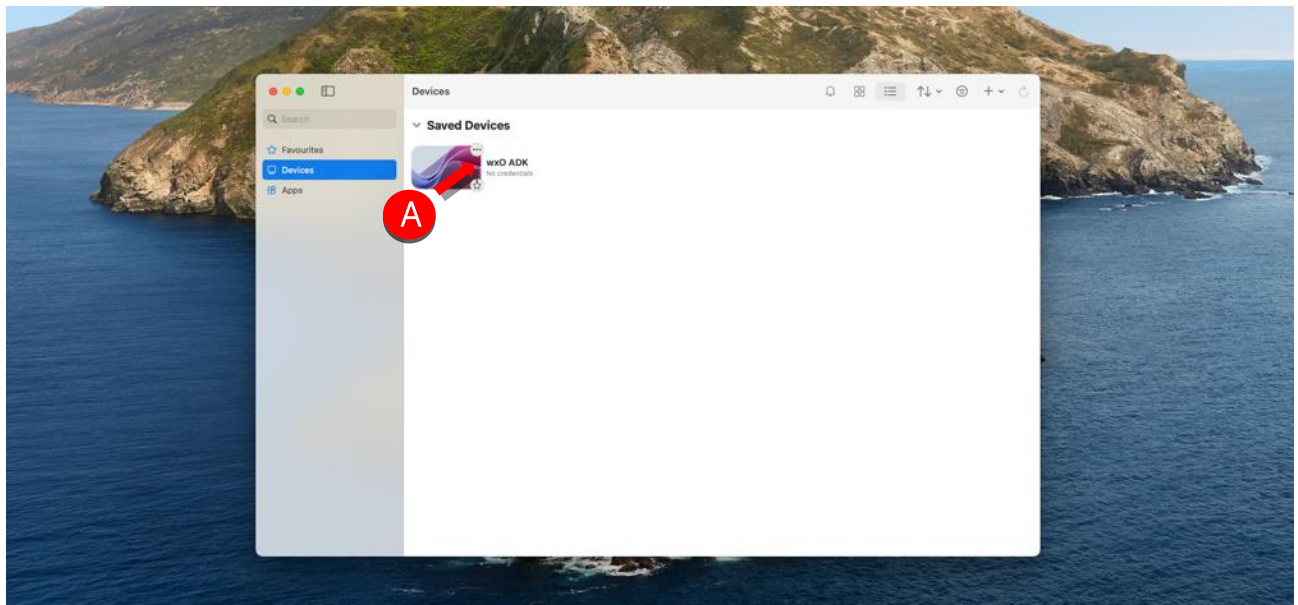
4. Enter “wx0 ADK” into the **Friendly name** field (A). Click **Add** (B).



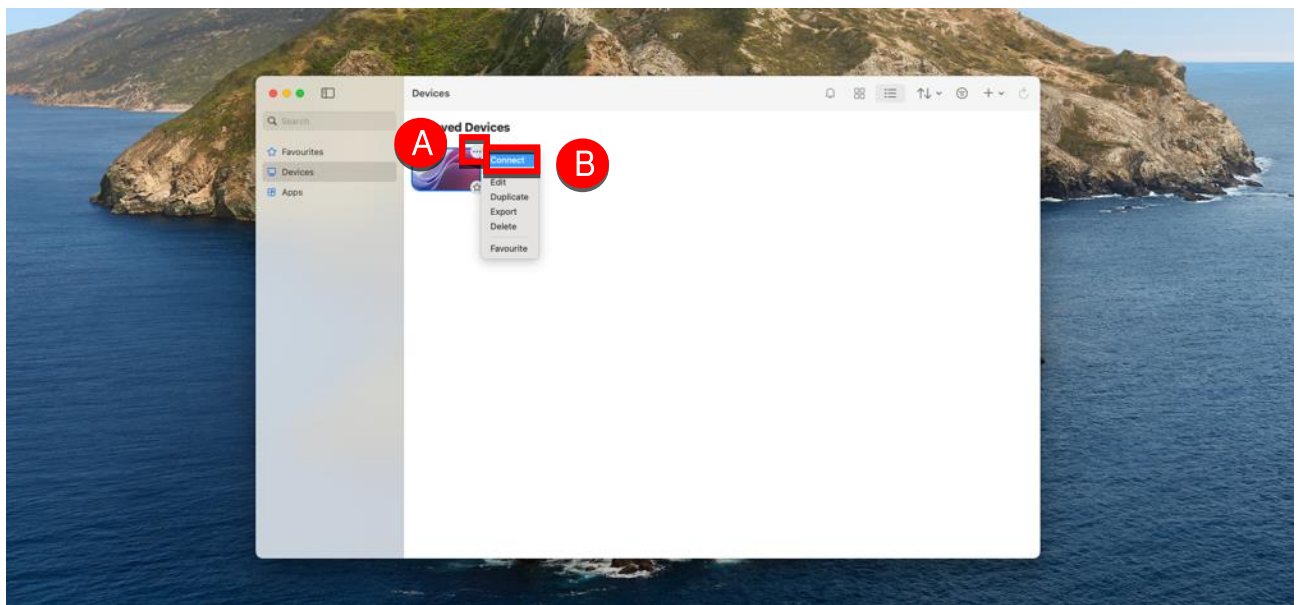
5. Click the **Devices** menu (A).



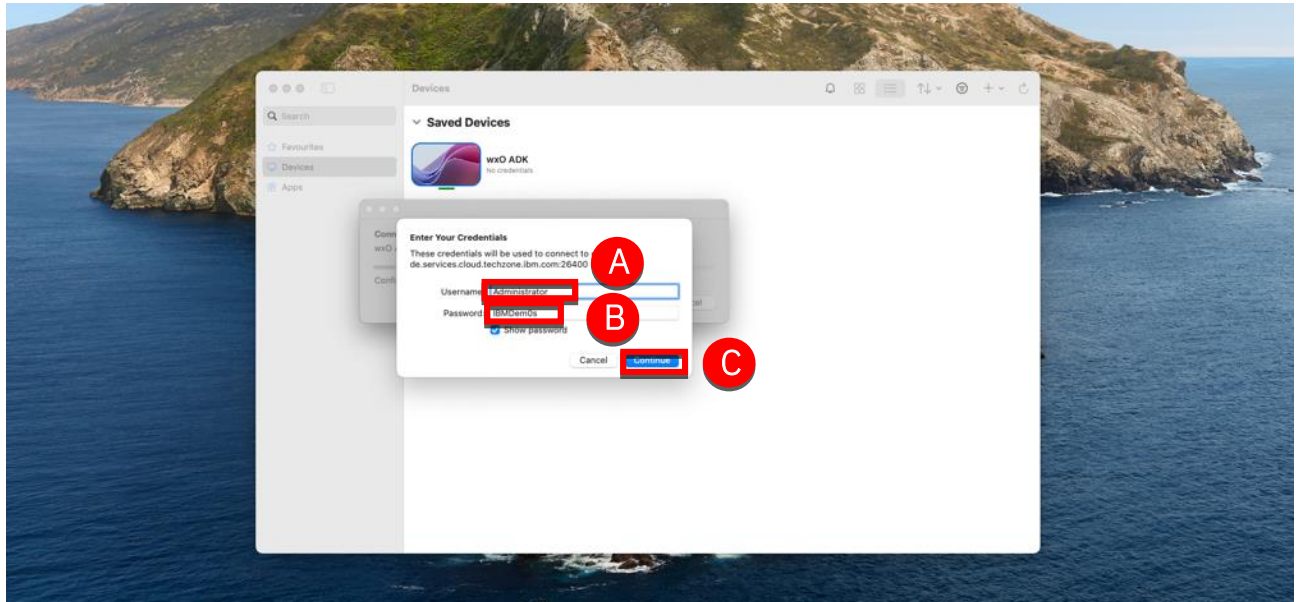
6. Place the cursor over your added PC to display the contextual menu (A).



7. Click the ... menu (A), then select **Connect** (B).

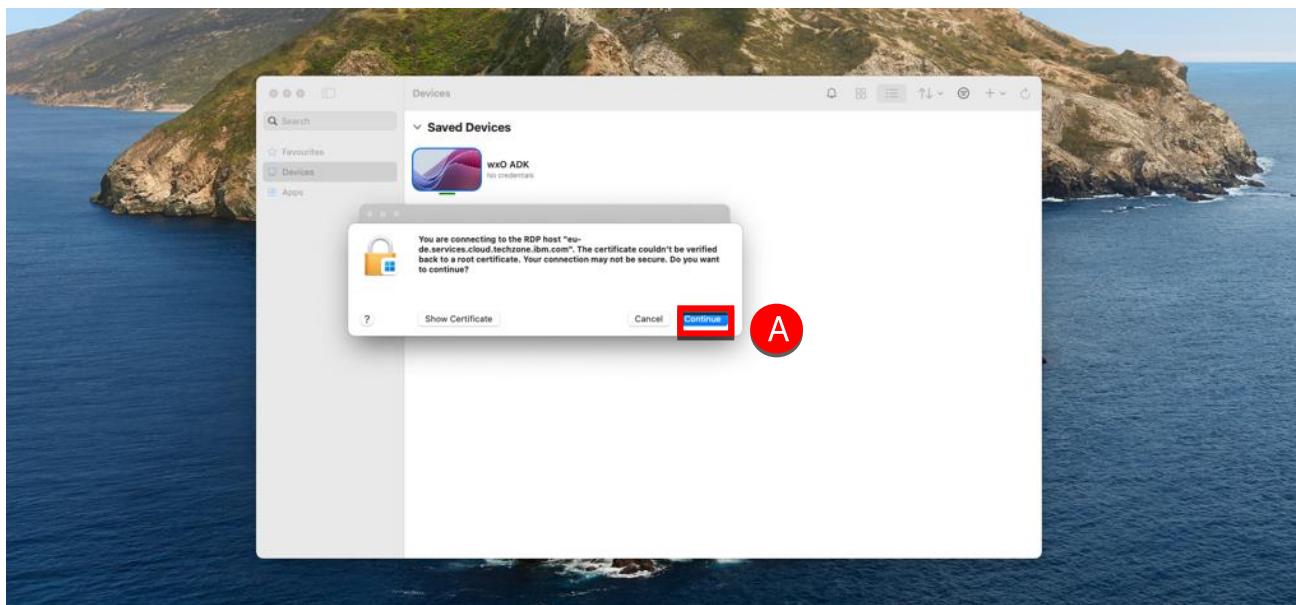


8. Type “Administrator” into **Username** (A). Type “IBMDemo0s” in **Password** (B). Click **Continue** (C).



Note: Some windows clients may need the “\” preceding “Administrator” (\Administrator) to indicate that this is an account on a remote machine, not on your local machine.

9. Click **Continue** (A).

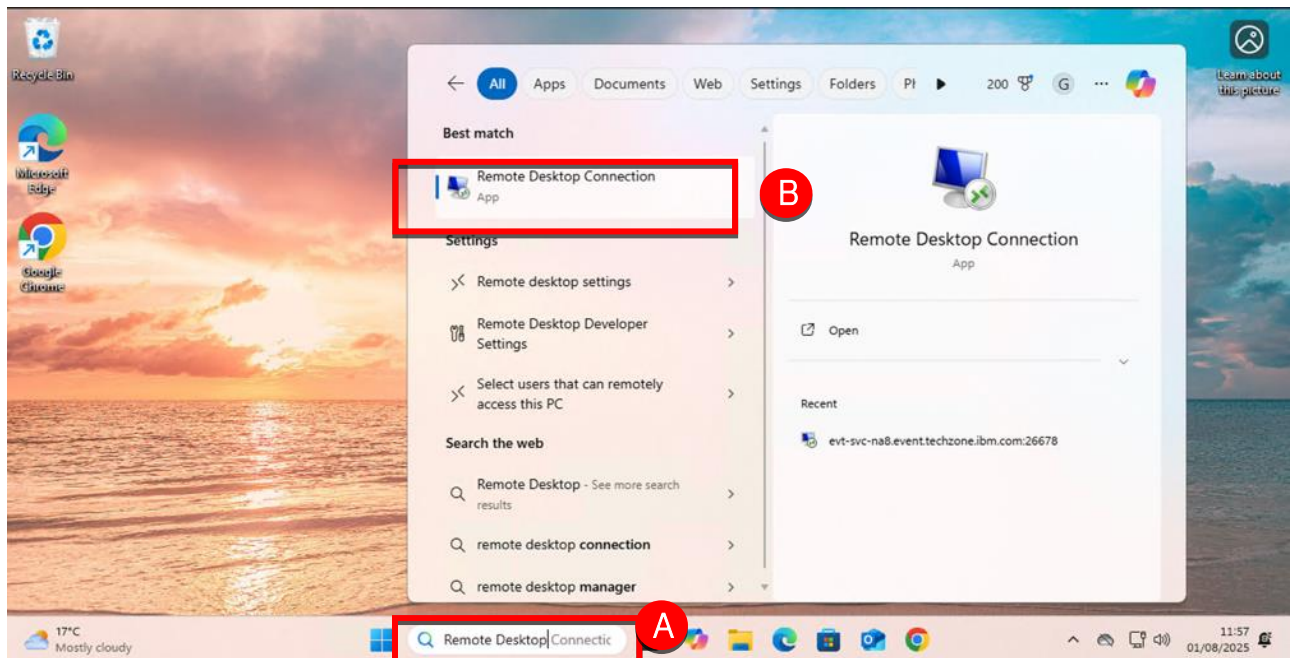


Access your image in Remote Desktop App (Windows Users)

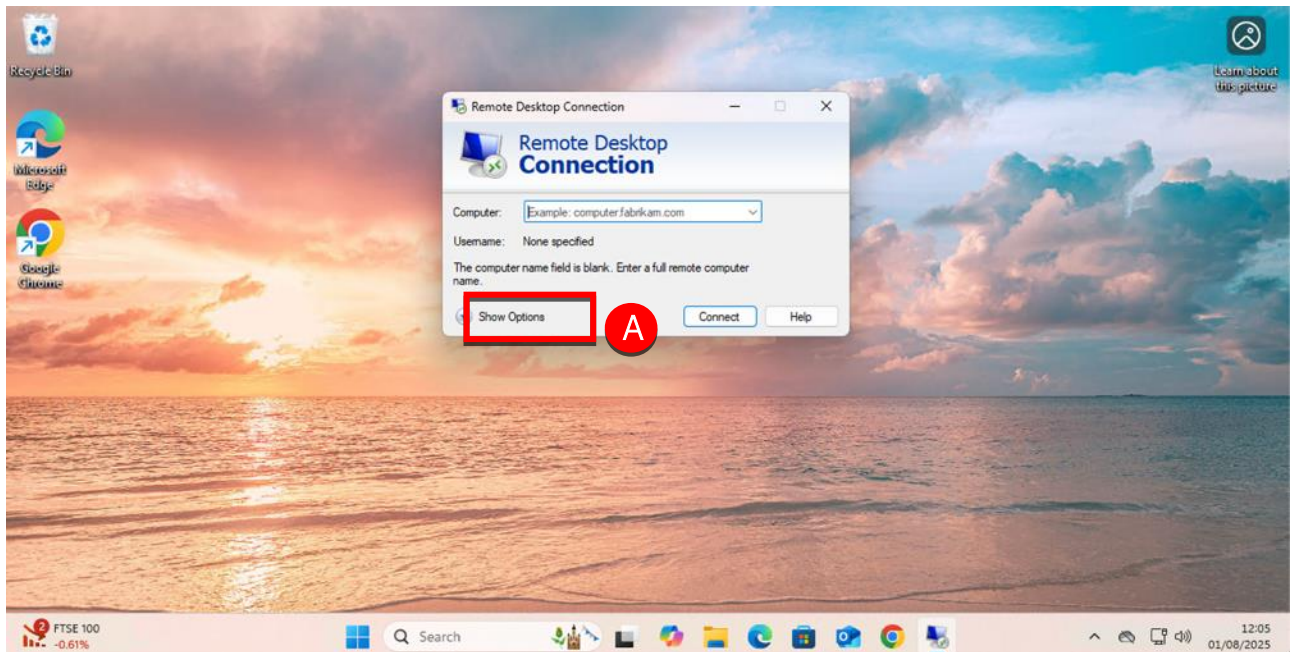
Skip this step if you have already connected to the remote machine from your Mac workstation. This section is specifically for Windows users connecting a Windows workstation to the remote machine.

If you will be connecting to the remote machine using a Windows workstation you will configure Remote Desktop Connection to access the VMWare image containing your pre-configured watsonx Orchestrate Developer Edition environment.

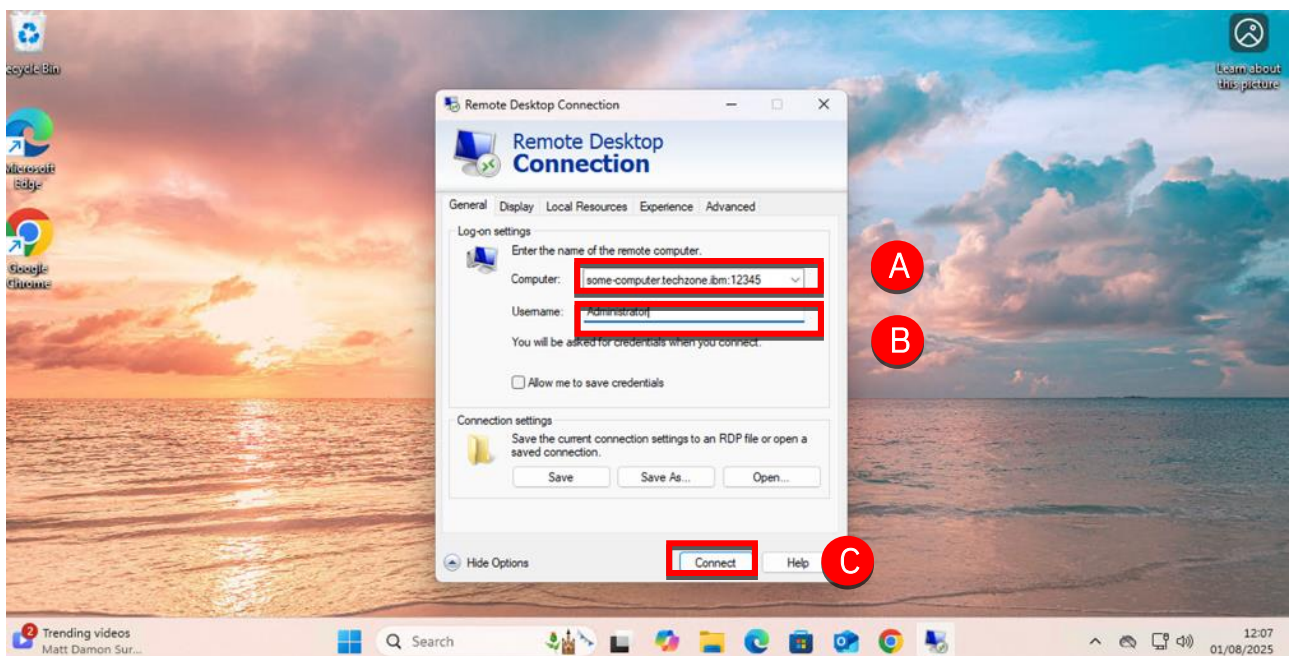
1. Start Remote Desktop Connection. Type “Remote Desktop” into the search bar (A), then click **Remote Desktop Connection** (B).



2. Click **Show Options** (A).

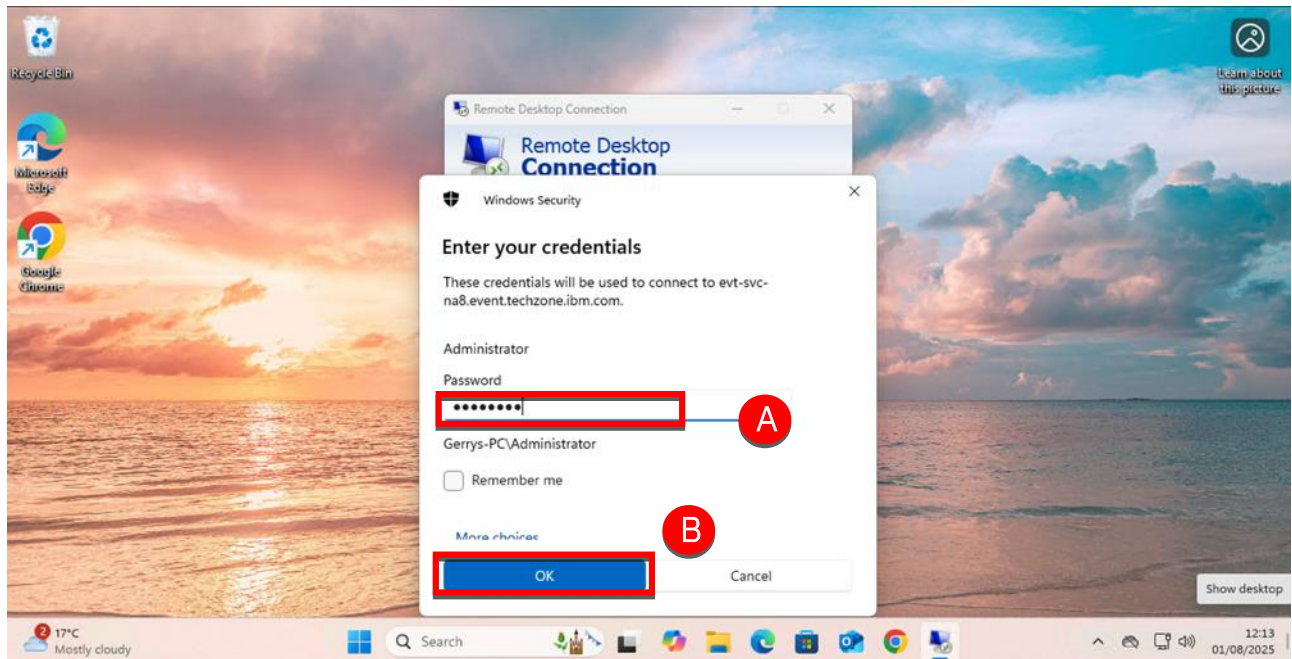


3. Enter the connection details for your reservation into the **Computer** field (Item 3) (A) and type "Administrator" into **Username** (B). Click **Connect** (C).

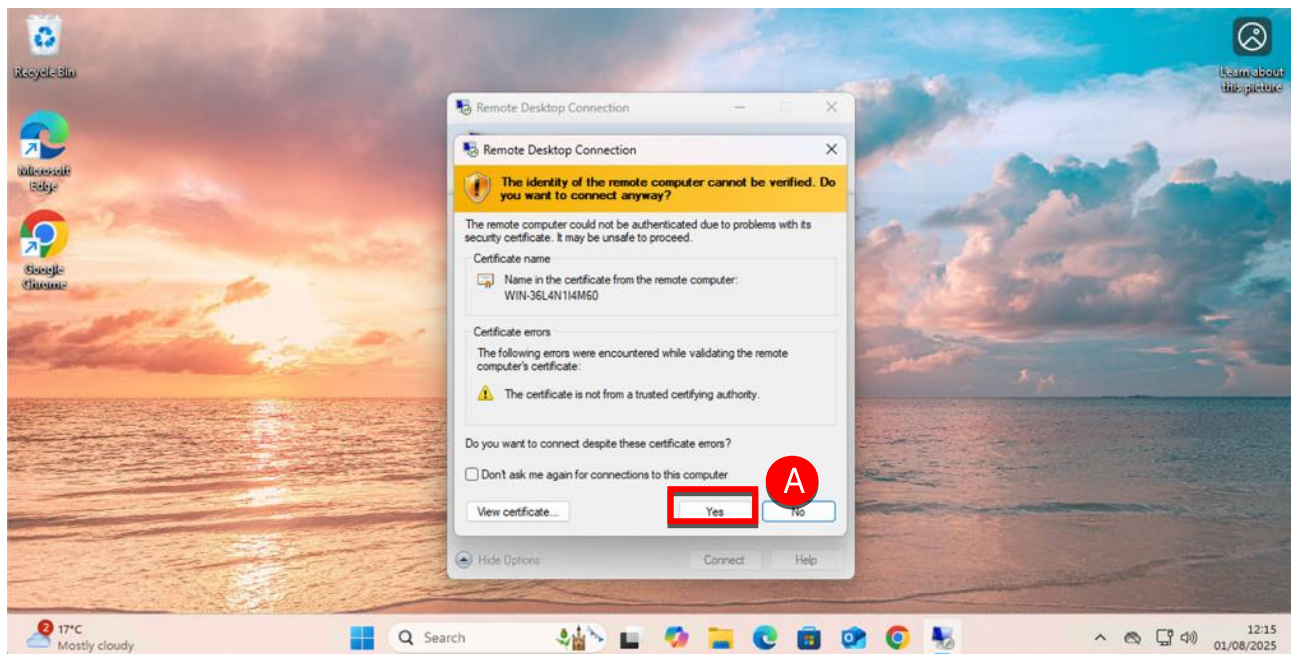


If you encounter a credentials error, use the username Administrator. If this also fails, try the username \\Administrator.

4. Type “IBMDem0s” in **Password** (A). Click **OK** (B).



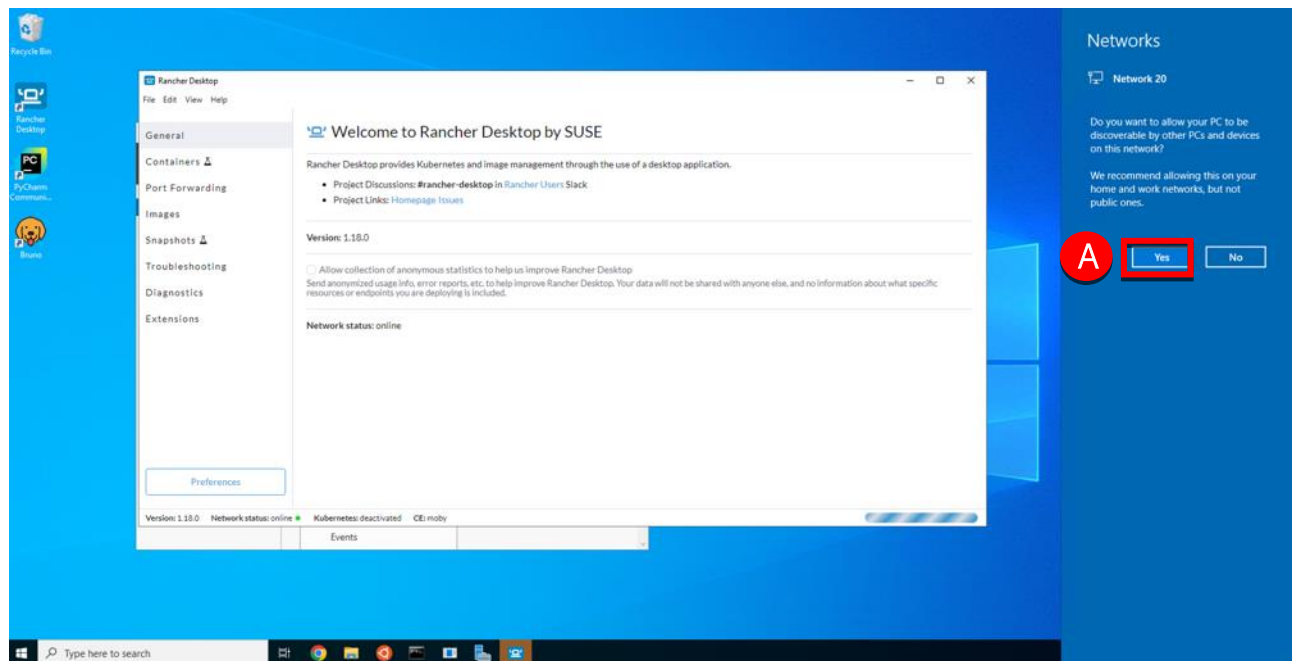
5. Click **Yes** (A) to connect, ignoring the certificate error.



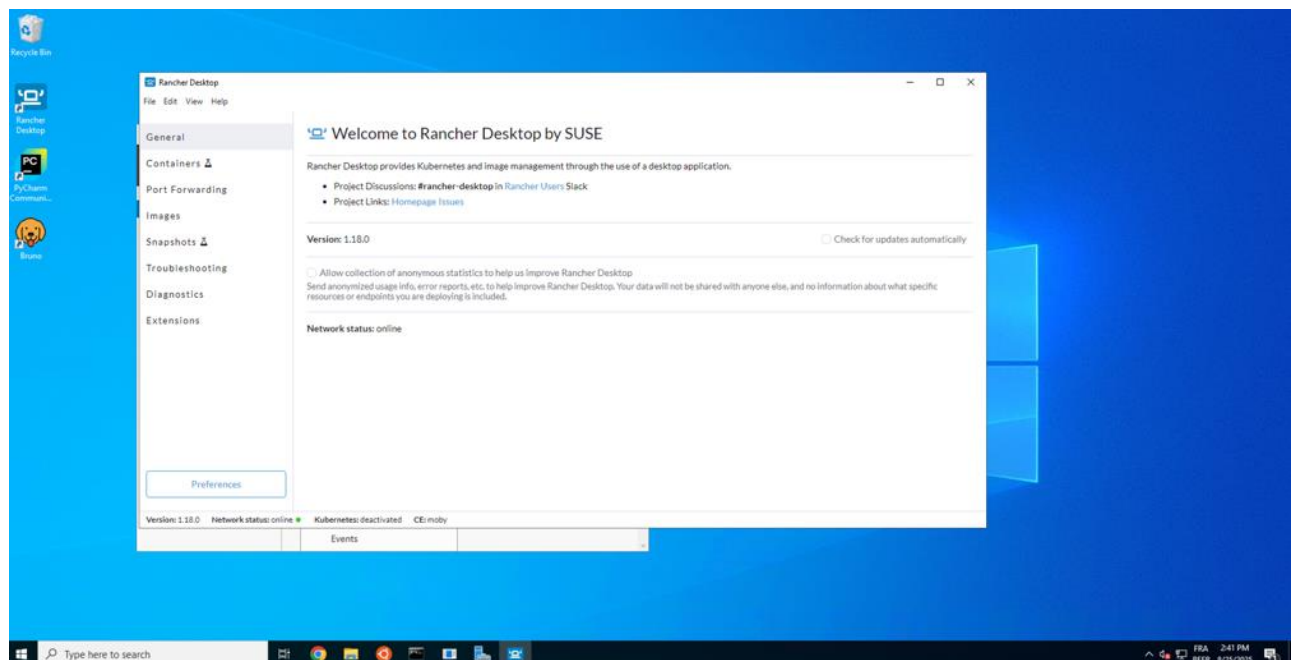
Application startup

It will take a few minutes for startup applications to be ready.

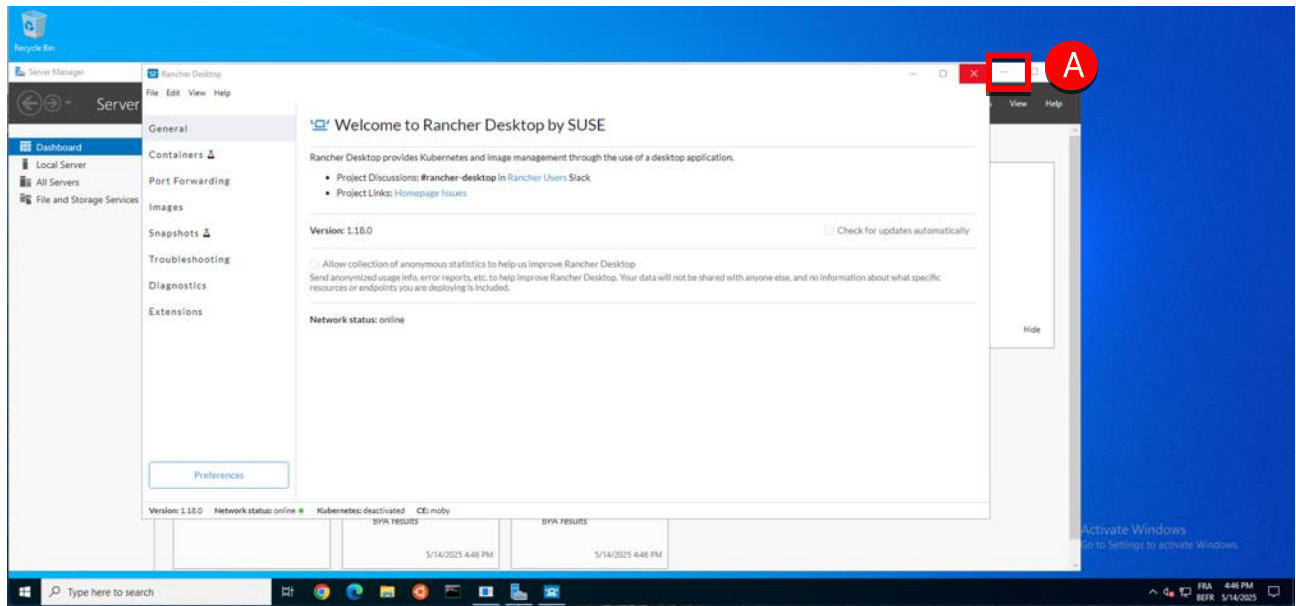
1. After a few minutes, your image should be started. In the **Networks** section, click Yes (A).



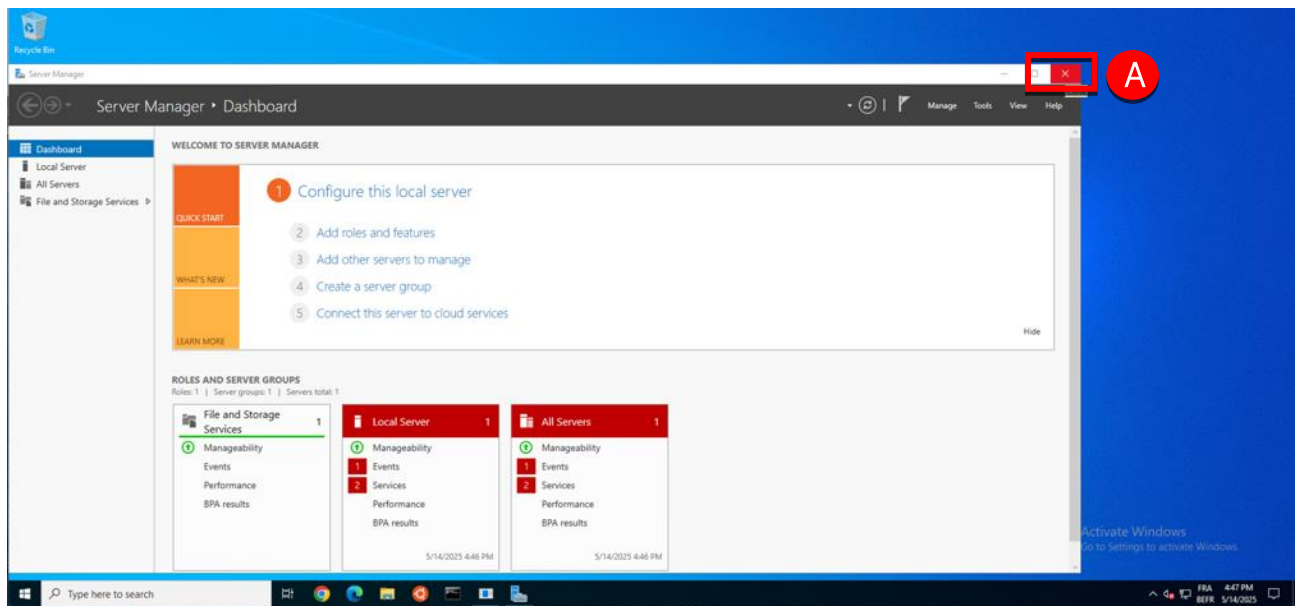
2. Wait for Rancher Desktop to be fully started (A) (no more activity listed).



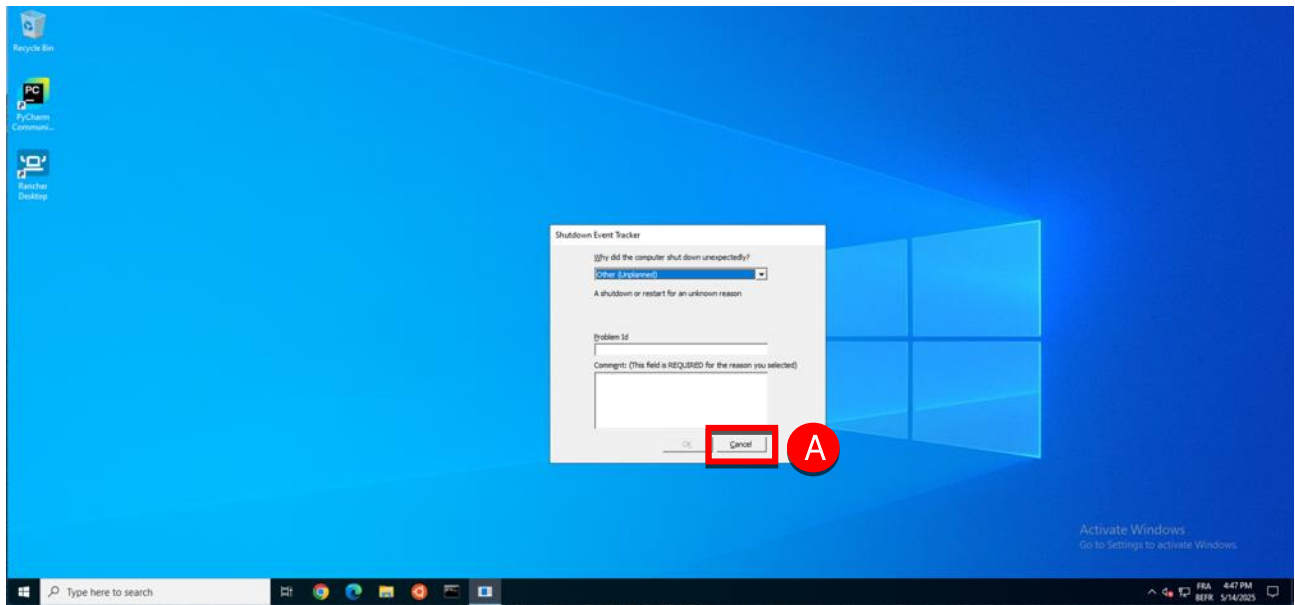
3. Click the icon to minimize the **Welcome Rancher Desktop** window (A). Note, even if closed the Rancher services will remain running.



4. Click X to close the **Server Manager** window (A)

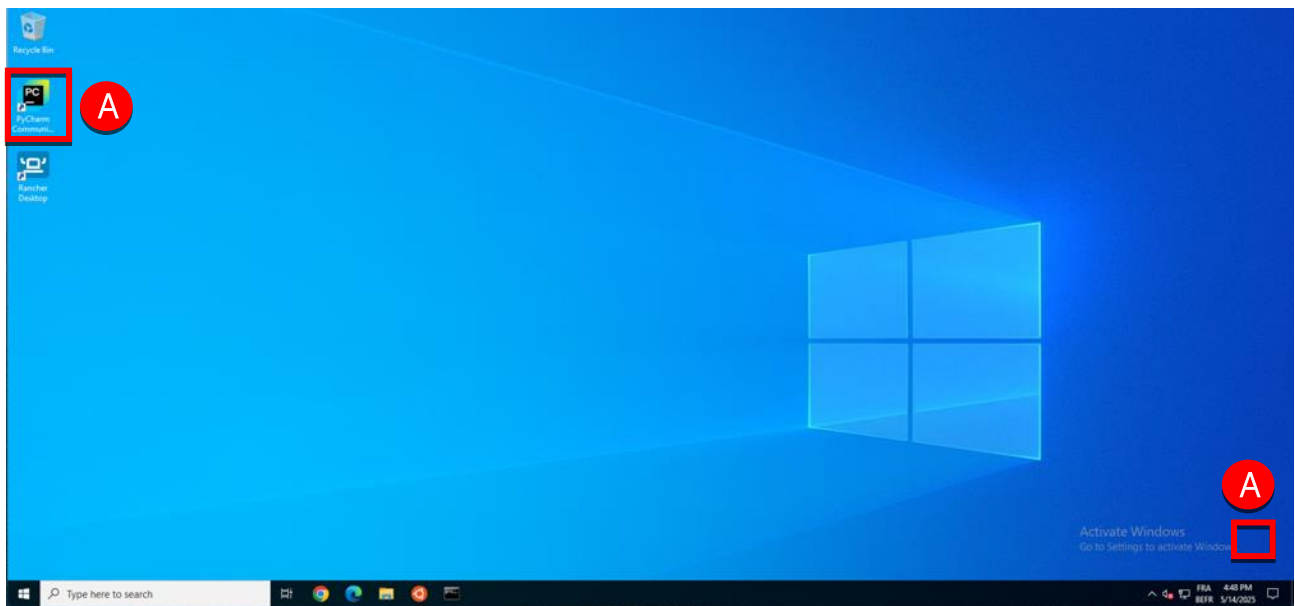


5. Click **Cancel** to close the Shutdown Event Tracker (A) (if required).

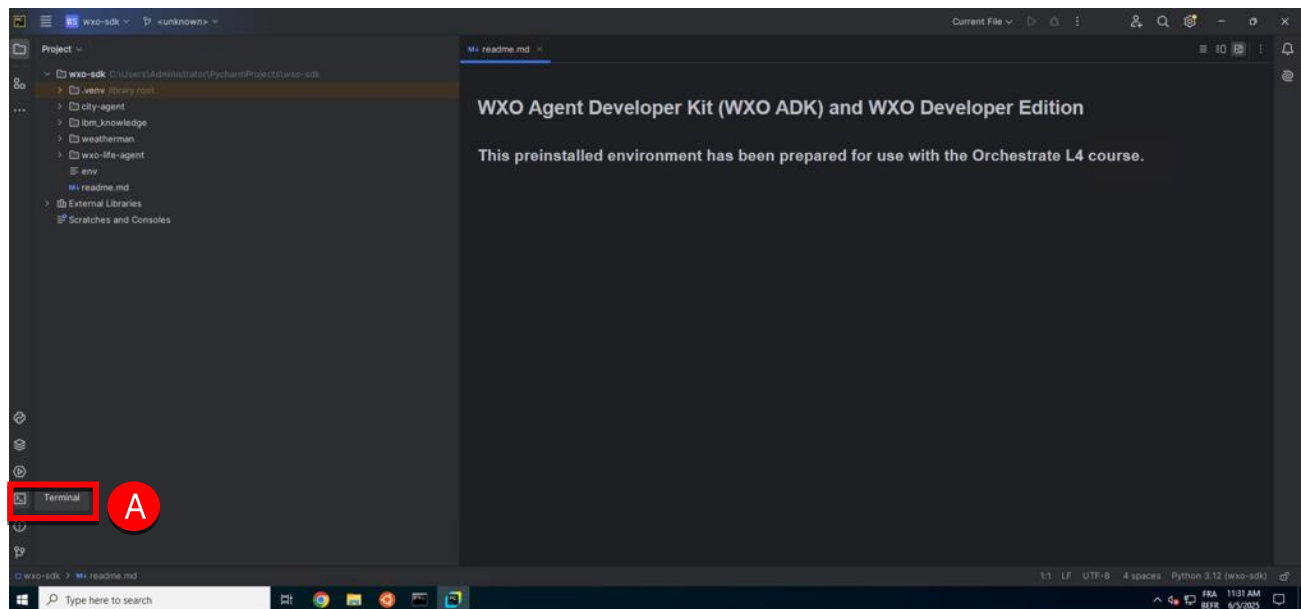


Configure your Orchestrate server

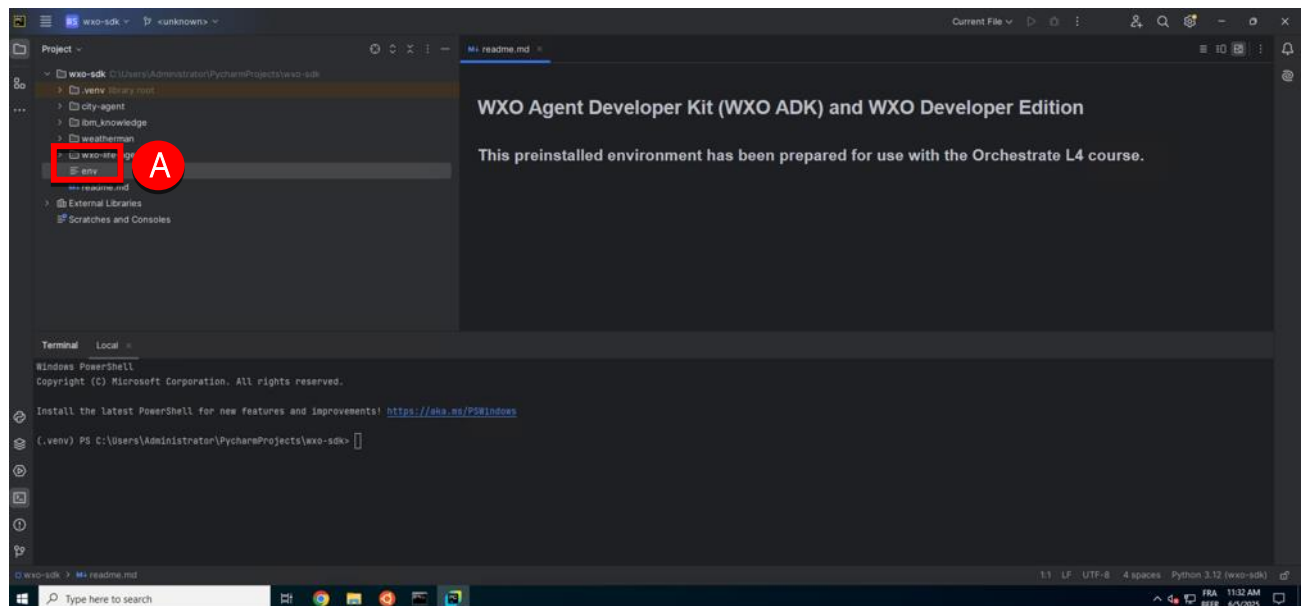
1. Double-click the **PyCharm** icon to open the **PyCharm Community edition** application.



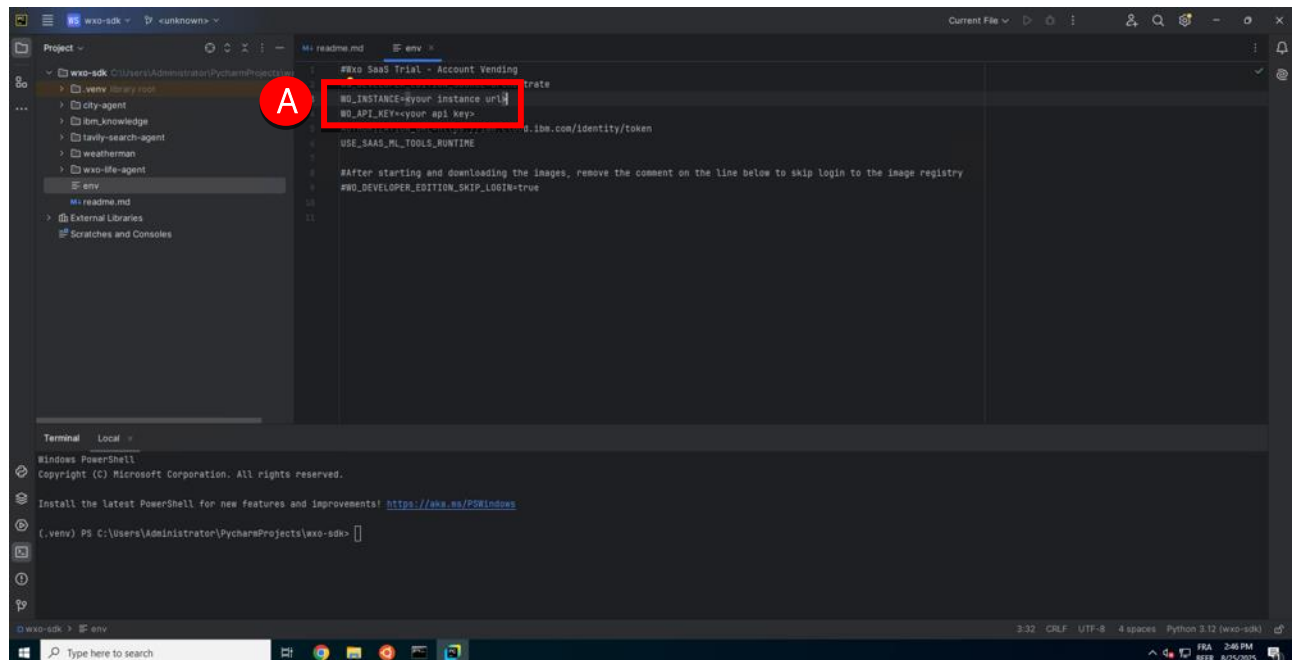
2. Click the **Terminal** icon to open the terminal panel (A) (if not already opened).



3. Double click the **env** file (A) in the **wso_sdk** folder to open the environment configuration.



The env file must contain the configuration parameters required to start the watsonx Orchestrate server. The parameters values (A) are the ones that have been collected and copied into your notes in the previous steps.

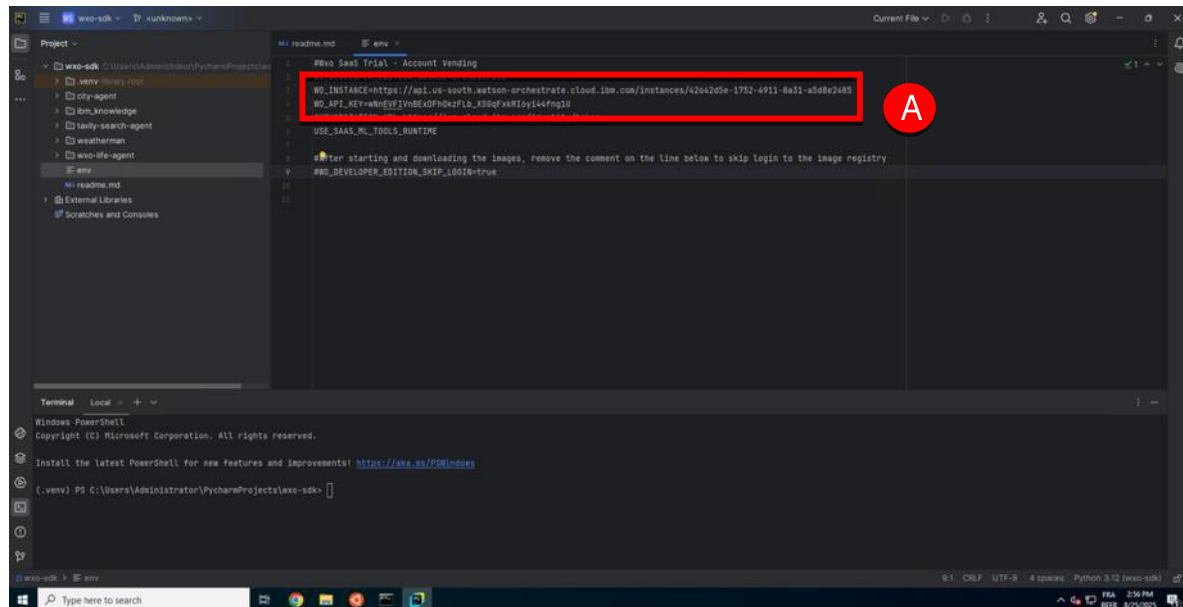


- **WO_INSTANCE** is the service URL you have collected from your SaaS environment ([get your instance URL](#)) and stored in item 7 in your note.
 - **WO_API_KEY** is the API key you have created in your IBM Cloud environment ([create your API key](#)) and stored in item 8 in your note.
4. Update the following text with your own keys (A) and copy the entire text to your clipboard.

```
WO_INSTANCE=<your SaaS service url>
WO_API_KET=<your IBM Cloud API key>
```

A

5. Replace the env file lines 3 and 4 with the text copied from the previous step (A).



```
Project - wso-sdk - <unknown>
  wso-sdk C:\Users\Administrator\PycharmProjects\wso-sdk
  venv library root
  city-agent
  km_knowledge
  taxi-search-agent
  weatherman
  wso-life-agent
  env
  wso-readme.md
  External Libraries
  Scratchers and Consoles

wso-readme.md
env
#wso-sdk Account Vending
#
# Instance ID: 4264205e-1732-4911-8a11-5d8e2485
# API Key: wmevyr1v8EXpH0czFl3_K8qPxxR1oy144fmg20
#
# USG_SAAS_MC_TOOLS_RUNTIME
#
# After starting and downloading the images, remove the comment on the line below to skip login to the image registry
#
#WS_DEVELOPER_EDITION_SKIP_LOGIN=true

Terminal Local - +
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

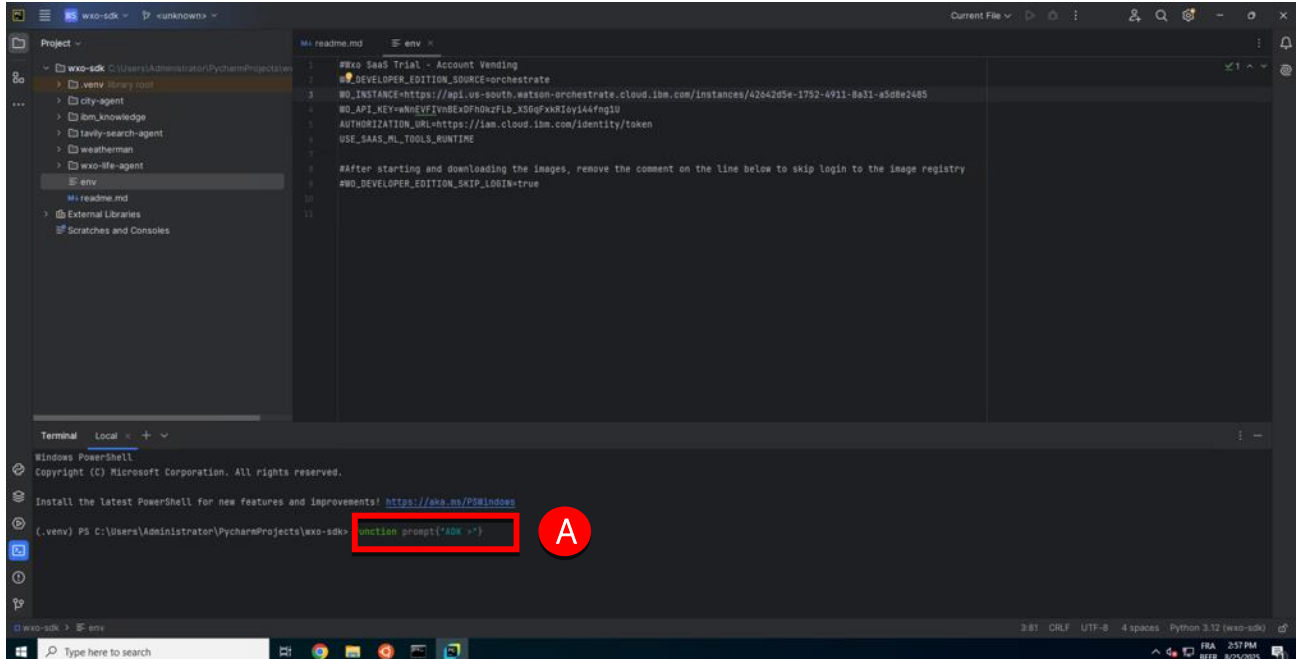
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

(.venv) PS C:\Users\Administrator\PycharmProjects\wso-sdk>
```

Let's now start the watsonx Orchestrate server.

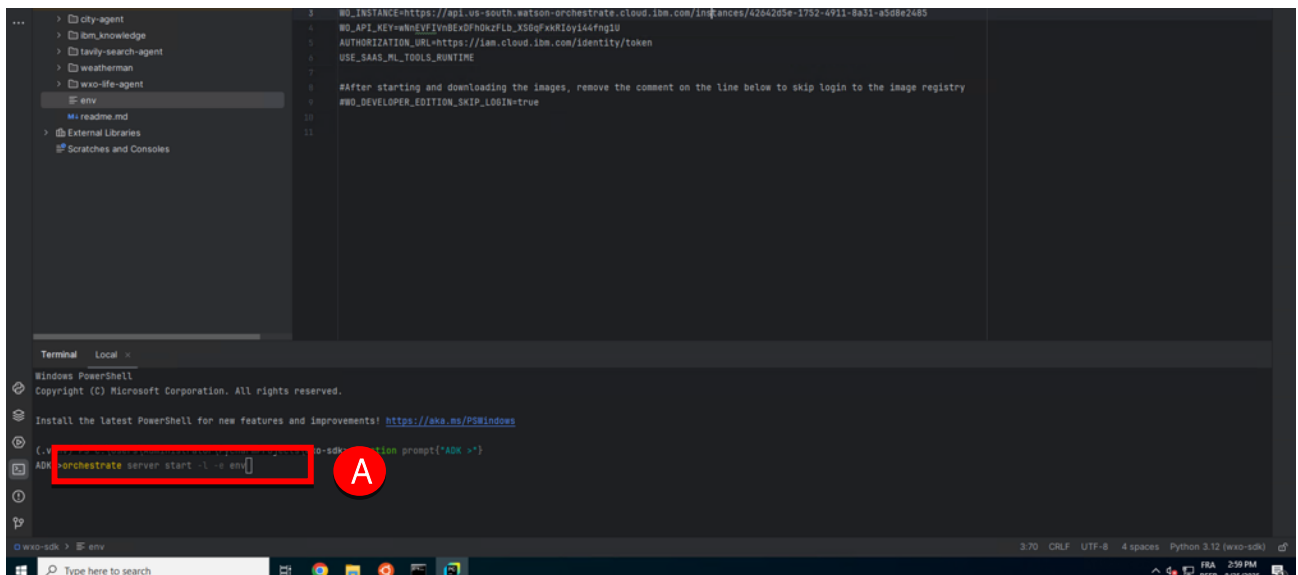
6. Enter the following command into the terminal to improve the command visibility, then press Enter:

```
function prompt{"ADK >"}
```



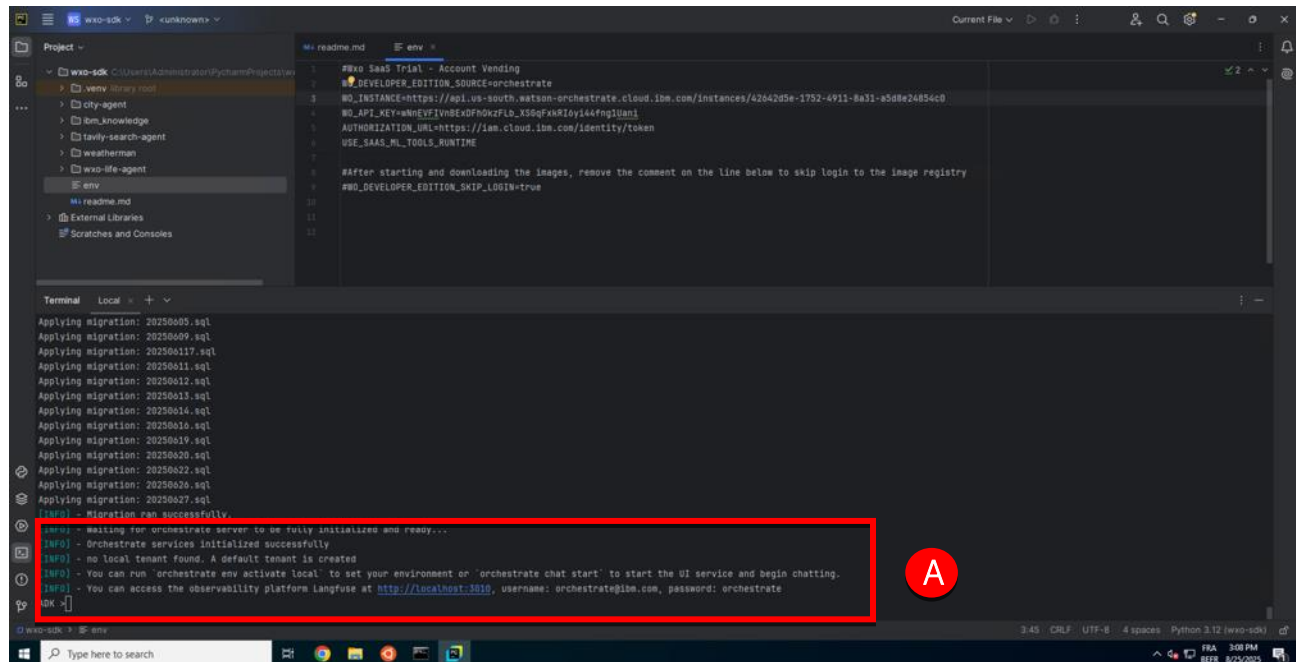
7. Enter the following command into the terminal to start the server with your env configuration, then press Enter.

```
orchestrate server start -l -e env
```



Note: Some services may fail to start on the first attempt, and you may encounter issues connecting to the docker repository. Retry the start command in this situation, depending on the volume of requests, it can take several attempts to connect.

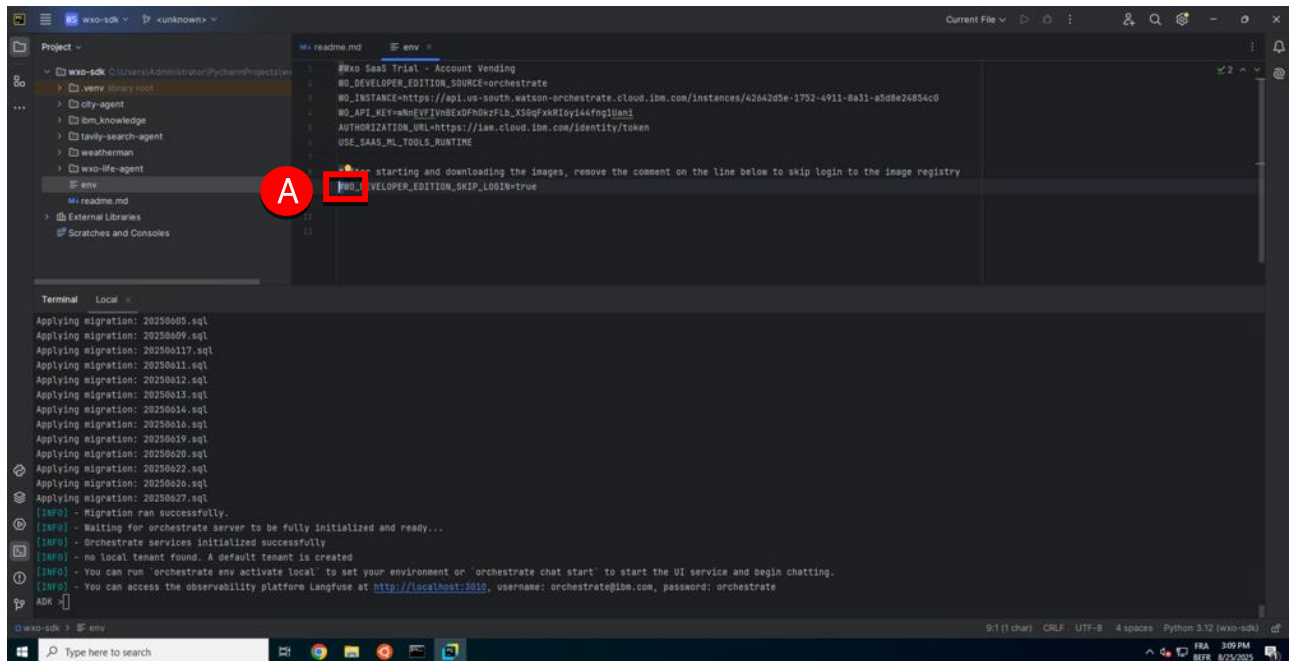
When startup is complete the terminal output will resemble the example below, restart can take 2-3 minutes to complete. The local server will be ready when you will see this **INFO** in the terminal screen (A):



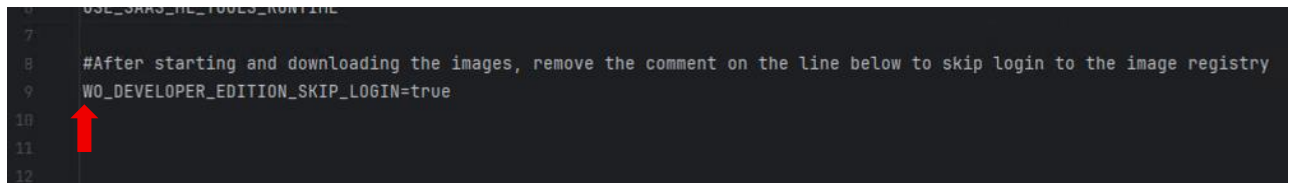
```
Project - wso-sdk - C:\Users\Administrator\PycharmProjects\wso-sdk
...
> .venv\library\tools
> city-agent
> iam-knowledge
> tavily-search-agent
> weatherman
> wso-lfe-agent
> .env
> External Libraries
> Scratches and Consoles

Terminal Local +
Applying migration: 20250605.sql
Applying migration: 20250609.sql
Applying migration: 202506117.sql
Applying migration: 20250611.sql
Applying migration: 20250612.sql
Applying migration: 20250613.sql
Applying migration: 20250614.sql
Applying migration: 20250616.sql
Applying migration: 20250619.sql
Applying migration: 20250620.sql
Applying migration: 20250622.sql
Applying migration: 20250626.sql
Applying migration: 20250627.sql
[INFO] - Migration ran successfully.
[INFO] - Waiting for orchestrator server to be fully initialized and ready...
[INFO] - Orchestrator services initialized successfully
[INFO] - no local tenant found. A default tenant is created
[INFO] - You can run 'orchestrator env activate local' to set your environment or 'orchestrator chat start' to start the UI service and begin chatting.
[INFO] - You can access the observability platform Langfuse at http://localhost:3030, username: orchestrator@ibm.com, password: orchestrator
OK -
```

8. Remove the comment (# character) from the env file as specified in the file instruction (A).

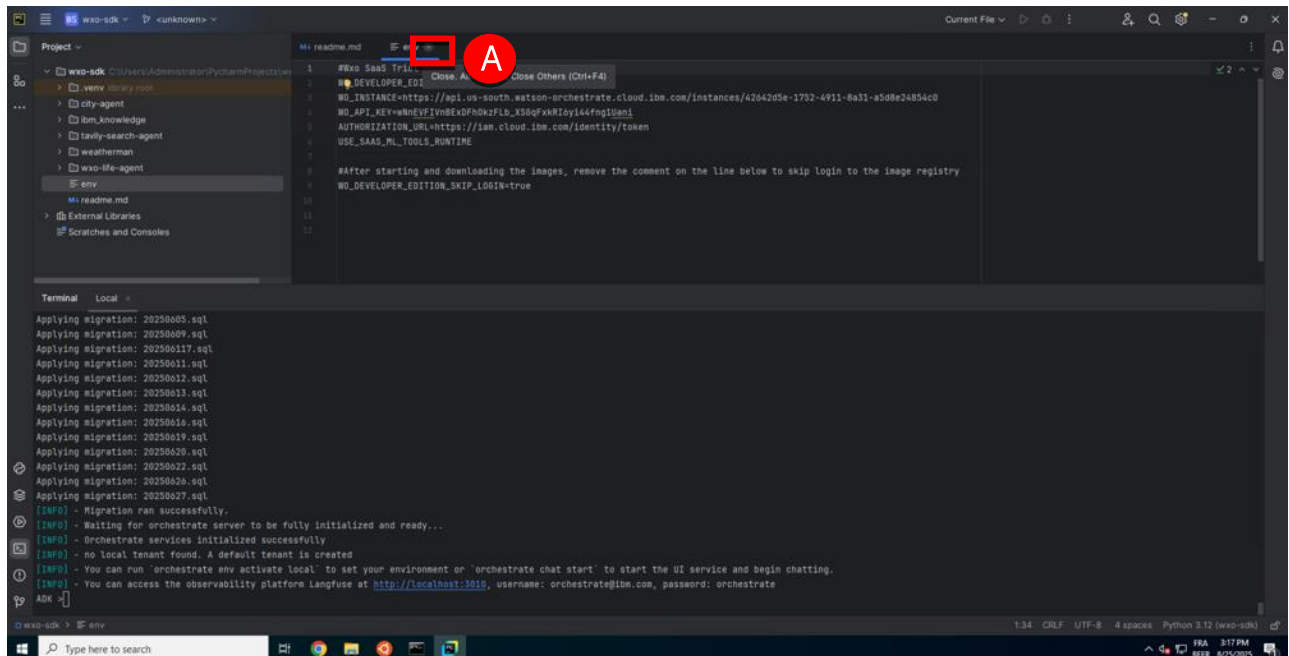


The last line in the env file should resemble the example below:



Now that the docker images have been downloaded, this setting skips the login into the image repository.

9. Click X to close the env file (A).



You are now ready to work with watsonx Orchestrate ADK.

Additional orchestrate ADK commands:

If you need to stop, or reset your server, or stop the chat server, the following commands can be used. Note that these steps are for information only, and do not need to be performed here.

If you need to stop your server, enter the following command into the terminal, then press Enter. (Don't do this if you have just setup your environment as you will need the server running).

```
orchestrate server stop
```

To reset your server and stop it, enter the following command into the terminal, then press Enter.

```
orchestrate server reset
```

To stop your chat service, enter the following command into the terminal, then press Enter.

```
orchestrate chat stop
```

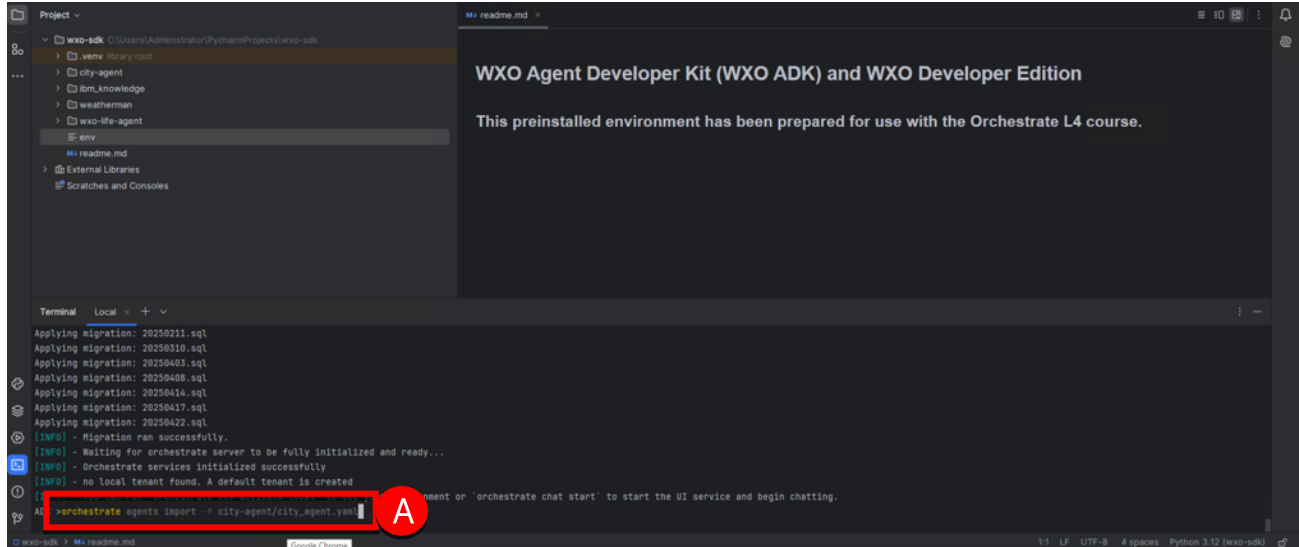
Validate your environment

In this section you will deploy a simple agent and test it using the watsonx Orchestrate chat interface. This step is just to validate that your environment is correctly configured to perform the other labs. The

configuration of the agent will not be explained in this section, it's only purpose is to validate that your environment has been configured correctly. These instructions assume your server is running.

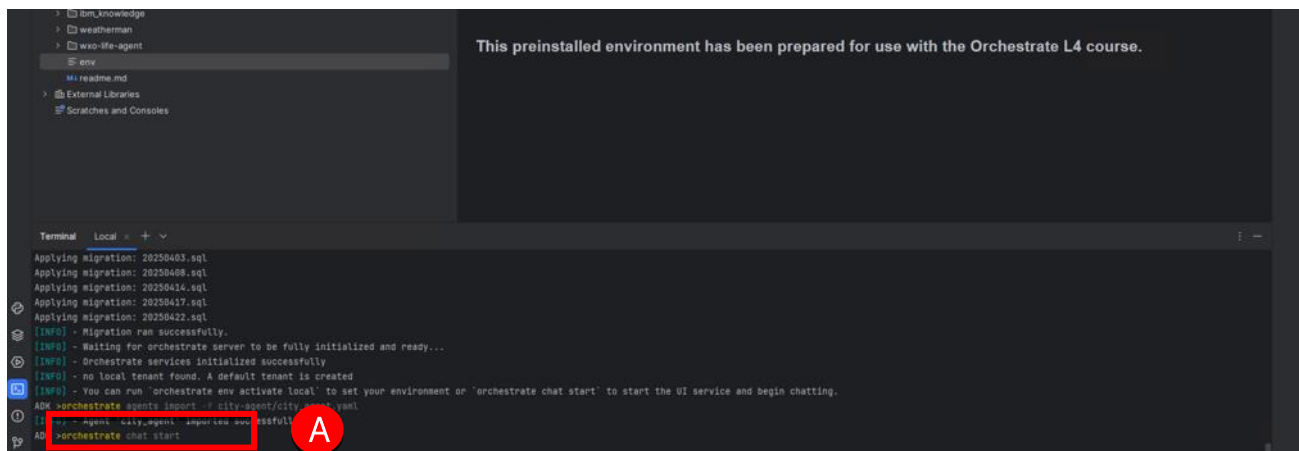
1. Enter the following command into the terminal (A) to create an agent.

```
orchestrate agents import -f city-agent/city_agent.yaml
```

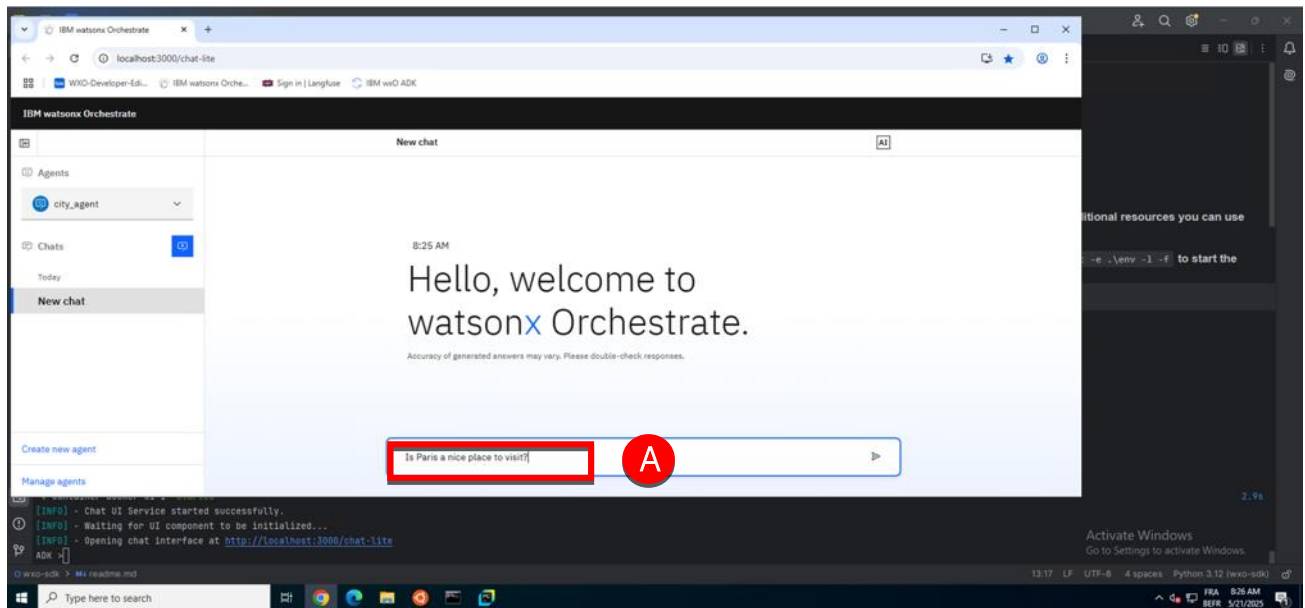


2. Enter the following command into the terminal (A) to start the chat server.

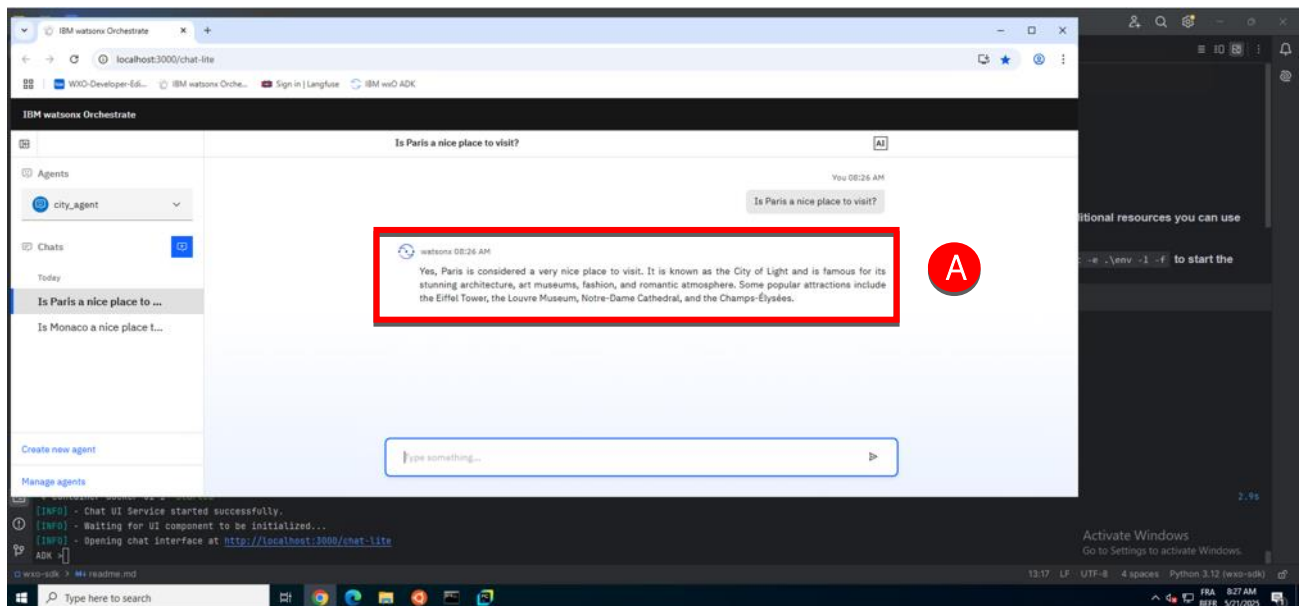
```
orchestrate chat start
```



- When the browser opens enter "Is Paris a nice place to visit?" into the chat window (A), then press Enter.



- Wait for the answer (A).



If the system responded with an answer, the agent is working correctly, and the system is configured correctly to work with the foundation models deployed in watsonx.ai.

Getting help

If you require assistance in interpreting any of the steps in this lab, please post your questions to the **#ba-techlcd-support** Slack channel (IBMer only).

Business Partners can request help at the [Partner Plus Support](#) website.

For troubleshooting tips, see the **TechZone Set Up Troubleshooting Guide** (<https://ibm.seismic.com/Link/Content/DCGT3pQ7hHM828WDQ86R7Tf6gpPV>) for help with common issues and solutions when using IBM Technology Zone. Additionally, go to the IBM Technology Zone Help page (<https://techzone.ibm.com/help>).

If you have an issue with the site, you can open a support case (https://ibmsf.force.com/ibminternalproducts/s/createrecord/NewCase?language=en_US). Alternatively, you can send an email to techzone.help@ibm.com. This contact information is also provided in the emails that you receive from IBM Technology Zone.

Additionally, see the watsonx.ai documentation (<https://dataplatfom.cloud.ibm.com/docs/content/wsi/getting-started/welcome-main.html?context=wx&audience=wdp>).

Or the watsonx Orchestrate DE documentation (https://developer.watson-orchestrate.ibm.com/getting_started/wxOde_setup)

Or the watsonx Orchestrate documentation (<https://www.ibm.com/docs/en/watsonx/watson-orchestrate/base?topic=getting-started-watsonx-orchestrate>)

